

● MEDIUM

● GEOMETRY AND TRIGONOMETRY

● ANSWER KEY

---

# Right triangles and trigonometry

# 02

4 answers · Rationale-rich · Teach from doc

**Q1****45**

The correct answer is 45. An isosceles right triangle has a right angle and two legs of equal length. In the triangle shown, one angle is a right angle and the two legs each have a length of 32. Thus, the given triangle is an isosceles right triangle. In an isosceles right triangle, the measures of the two non-right angles are  $45^\circ$ . It follows that the value of  $x$  is 45.

**Q2****B**

B.  $\frac{308}{317}$

Choice B is correct. If two triangles are similar, then their corresponding angles are congruent. It's given that right triangle  $FGH$  is similar to right triangle  $JKL$  and angle  $F$  corresponds to angle  $J$ . It follows that angle  $F$  is congruent to angle  $J$  and, therefore, the measure of angle  $F$  is equal to the measure of angle  $J$ . The sine ratios of angles of equal measure are equal. Since the measure of angle  $F$  is equal to the measure of angle  $J$ ,  $\sin F = \sin J$ . It's given that  $\sin F = \frac{308}{317}$ . Therefore,  $\sin J$  is  $\frac{308}{317}$ .

Choice A is incorrect. This is the value of  $\cos J$ , not the value of  $\sin J$ .

Choice C is incorrect. This is the reciprocal of the value of  $\sin J$ , not the value of  $\sin J$ .

Choice D is incorrect. This is the reciprocal of the value of  $\cos J$ , not the value of  $\sin J$ .

### Q3

B

B. 6

Choice B is correct. Triangles ADB and CDB are both  $30^\circ - 60^\circ - 90^\circ$  triangles and share  $\overline{BD}$ . Therefore, triangles ADB and CDB are congruent by the angle-side-angle postulate. Using the properties of  $30^\circ - 60^\circ - 90^\circ$  triangles, the length of  $\overline{AD}$  is half the length of hypotenuse  $\overline{AB}$ . Since the triangles are congruent,  $AB = BC = 12$ . So the length of  $\overline{AD}$  is  $\frac{12}{2} = 6$ .

Alternate approach: Since angle CBD has a measure of  $30^\circ$ , angle ABC must have a measure of  $60^\circ$ . It follows that triangle ABC is equilateral, so side AC also has length 12. It also follows that the altitude BD is also a median, and therefore the length of AD is half of the length of AC, which is 6.

Choice A is incorrect. If the length of  $\overline{AD}$  were 4, then the length of  $\overline{AB}$  would be 8. However, this is incorrect because  $\overline{AB}$  is congruent to  $\overline{BC}$ , which has a length of 12. Choices C and D are also incorrect. Following the same procedures as used to test choice A gives  $\overline{AB}$  a length of  $12\sqrt{2}$  for choice C and  $12\sqrt{3}$  for choice D. However, these results cannot be true because  $\overline{AB}$  is congruent to  $\overline{BC}$ , which has a length of 12.

**Q4****A**

A.  $\sqrt{21^2 - 6^2}$

---

Choice A is correct. For the right triangle shown, the lengths of the legs are  $a$  units and 6 units, and the length of the hypotenuse is 21 units. The Pythagorean theorem states that in a right triangle, the sum of the squares of the lengths of the two legs is equal to the square of the length of the hypotenuse. Therefore,  $a^2 + 6^2 = 21^2$ . Subtracting  $6^2$  from both sides of this equation yields  $a^2 = 21^2 - 6^2$ . Taking the square root of both sides of this equation yields  $a = \pm \sqrt{21^2 - 6^2}$ . Since  $a$  is a length,  $a$  must be positive. Therefore,  $a = \sqrt{21^2 - 6^2}$ . Thus, for the triangle shown,  $\sqrt{21^2 - 6^2}$  represents the value of  $a$ .

Choice B is incorrect. For the triangle shown, this expression represents the value of  $a^2$ , not  $a$ .

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

---

#### Notes

Accept equivalent forms (e.g.  $0.25 \equiv 1/4$ ). For grid-in items, decimal and fraction answers are both valid.

Generated 2026-05-29 · cb-educator-question-bank